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HIBISCUS ROSA-SINENSIS: A BOTANICAL MARVEL WITH MULTIFARIOUS PHYSIOLOGICAL ACTIVITIES

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ABSTRACT

Hibiscus rosa-sinensis L., commonly known as China rose, is a medicinal plant with a rich history of traditional use. This comprehensive review explores the diverse pharmacological properties of Hibiscus rosa-sinensis, including its antibacterial, antifungal, antidiabetic, and anticancer activities. Additionally, it exhibits antioxidant, anti-inflammatory, and wound healing effects, making it a valuable botanical resource for various healthcare needs. From its historical significance in ancient texts to its integration into contemporary Ayurvedic databases, Hibiscus rosa-sinensis holds promising potential for modern medicine and natural remedies.

KEYWORDS: Hibiscus rosa-sinensis, China rose, medicinal plant, pharmacological properties, modern medicine, natural remedies.

INTRODUCTION

Throughout history, medicinal plants have been extensively utilized for various purposes due to their remarkable ability to synthesize chemical compounds that play a vital role in preventing numerous diseases, including cancer and diabetes. These plants boast a diverse array of chemical compounds, each serving essential biological functions and acting as a defense mechanism against predators such as fungi, insects, and herbivorous mammals. To date, scientists have isolated approximately 12,000 of these compounds, although this figure is believed to represent less than 10% of the total number of compounds present.[1,2] The potential of medicinal plants and their chemical constituents continues to be a subject of great interest and exploration in the field of medicine and natural remedies.

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Medicinal plants can be classified into three significant sectors based on their usage.

(a) Modern medicines rely on approximately 30 to 35 medicinal plant species.

(b) Organized and codified traditional medicines, such as Ayurveda, Siddha, Unani, Amchi, and Tibetan systems, draw from a broader range of about 1,200 to 2,000 medicinal plant species.

(c) Local health traditions encompass practices followed by villagers, folk healers, vaidyas, and tribal communities, utilizing an extensive array of over 8,000 species of medicinal plants primarily for essential healthcare needs.[3]

Each of these sectors represents a unique approach to harnessing the therapeutic potential of medicinal plants, contributing to a diverse and rich landscape of traditional and modern healthcare systems.

The Hibiscus rosa sinensis L. (Malvaceae) herb is originally native to China and is well-known for its stunning, showy flowers. This shrub is widely cultivated in tropical regions as an ornamental plant, presenting various forms with flowers of different colors. Beyond its aesthetic appeal, Hibiscus also possesses medicinal properties, making it a key ingredient in numerous herbal teas.

Among the different varieties, the red-flowered variety is particularly favored for its medicinal applications. Numerous studies have reported that different varieties of Hibiscus plants exhibit varying medicinal properties. However, this review primarily centers on exploring the therapeutic potential of the Hibiscus rosa sinensis plant and its wide-ranging applications in traditional and modern medicine.

MATERIALS AND METHODS

This paper includes an evidence-based overview of the pharmacological properties and toxicity studies of the Hibiscus rosa-sinensis, which may be helpful to establish a standard natural drug for further research. This article presents an in-depth review of Japa (Hibiscus) from different classical texts, spanning various historical periods. The present article provide review on Hibiscus rosa-sinensis Linn. Collecting information on the various pharmacological studies conducted till date.

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Etymology Of The Word[4]

The word "Hibiscus" has interesting origins: Dioscorides associated it with the Marshmallow plant. "Hibiscus" refers to a plant belonging to the mallow family, thriving in warm climate tes, known for its striking, large, and brightly colored flowers. It also has applications in producing fibers or timber.

The term "rosa-sinensis" was attributed to the deep-red flowered variety, believed to have originated in Asia, hence the name.

Chronological Observations

Vedic Period[5]

The Hibiscus flowers find special use in Hindu deity worship, particularly in the reverence of Goddess Durga/Kali. The glowing complexion of the Devi is often compared to the beauty of the Hibiscus flower in the Lalitopaakhyaana. The Sun God is also associated with the Hibiscus flower, as seen in a well-known shloka that begins with 'Japa kusumasankarsham, kashyapeyammaha-dyuthim'.

Samhita Period

No references to Japa are found in texts from this period.

Nighantu Kala[6-8]

Several Nighantus, including Amarakosha, Paryayaratnamala, and Raja Nighantu, mention Japa.

Raja Nighantu classifies it under karavira. Bhavaprakash Nighantu includes Japa in the Pushpavarga. Amarakosha refers to it as java in the Vanaushadhivarga.

Adhunika Kala

Japa is identified as Hibiscus rosa-sinensis Linn. in contemporary Ayurvedic databases, such as the Ayurvedic Pharmacopoeia of India.

Detailed information on its Morphological features, Vernacular names, Microscopic features, Phytochemicals, and Therapeutic uses is available. Japa is listed in the karpasadi varga of Nighantu Adarsha. Pushpa Ayurveda explains its therapeutic uses. Various regional books, including Paramparika Vaidyarachikitsa, Amahitidravyakosha in Karnataka, Sarvarogachikitsaratnam in Kerala, and Flora of Shimoga, document folklore practices related to Japa.

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Taxonomical Classification:

Kingdom : Plantae
Subkingdom : Tracheobionta
Super Division : Spermatophyta
Division : Magnoliophyta
Class : Magnoliopsida
Subclass : Dilleniidae
Order : Malvales
Family : Malvaceae
Genus : Hibiscus
Species : Hibiscus rosa-sinensis[9]

Common Names

China rose, China rose plant, Gudhal, Gurhal, Jaba, Mandaar, Japaphool, Jasum, Jasunt, Jaswand, Jiapushpa.

Vernacular Names

English : Shoe-flower plant, Chinese Hibiscus.
Hindi : Jasut, Jasum, Java, Odhul, Gurhal, Arahul.
Marathi : Jasavanda, Jassvandi. Sanskrita- Japa, Java, Rudrapuspa, Aundrapuspa, Trisandhya.
Bengali : Joba, Jiwa, Oru.
Gujarati : Jasvua, Jasunt.
Kannada : Dasavala.
Malayalam : Himbarathi, Ayamparatti, Chebarathi.
Punjabi : Jasum, Jaipushpa, Gurhal.
Tamil : Sapattuu, Semparutti.[10]

Rasa Panchaka

Rasa : Kashaya, Tikta, Madhura.
Guna : Laghu, Snigdha.
Veerya : Sheeta.
Vipaka : Katu, Madhura.
Doshaghnata : Kaphapittashamaka.

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Rogaghnata : Kshata, Indralupta, Daha, Unmada, Kasa.

Karma : Keshya, Hirdya, Raktarodhaka, Stambhana[7]

Economic And Botanical Significance of Hibiscus

Shoe Flower: The dried powder of Hibiscus is utilized as shoe shine or polish, hence earning it the name "shoe flower." Moreover, its dry flower powder is a common component in various herbal tea blends.

Economic Importance: The mallow family holds considerable economic significance due to the abundance of fiber-yielding plants. Nearly all genera within this family possess some form of fiber-producing capability.

Dye Production: Petals of Hibiscus are used to create dyes, adding to their practical applications.

Landscaping: Hibiscus plants are often employed for hedges and screens, enhancing their role in landscaping.

Cosmetics: The juice extracted from Hibiscus petals is utilized in China for shoe-blacking and mascaras, showcasing its diverse applications.

Substitutes And Adulterants

Adulteration: The drug powder of Hibiscus may sometimes be adulterated with H.schizopetalus Hook.f., H. mutabilis Linn., and Malvaviscusmollis DC. To distinguish the genuine drug, one can observe the presence of various types of trichomes.

Culinary Use: Young leaves of Hibiscus occasionally serve as a substitute for spinach in culinary preparations.

Parts Used And Dosage

Parts Used : Flowers, leaves, and roots.

Dosage : Typically, the powder is administered in quantities ranging from 5 to 10 grams.

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Geographical distribution

The origin and historical distribution of the red Hibiscus remain a mystery. The genus Hibiscus comprises around 200 species, which are naturally found across tropical and subtropical regions. As a popular choice, Hibiscus is often cultivated as a garden ornamental, thriving in elevations from sea level up to 500 meters. However, it is sensitive to frost and may experience freezing in mild winters. Nonetheless, it exhibits resilience by regenerating from the base during spring. To ensure optimal growth, it flourishes in well-drained, fertile soil with abundant organic matter, basking in full sunlight.

Pharmacological/ Biological Activities

1. Antibacterial Activity

Methanol extracts from H. rosa-sinensis leaves exhibited antibacterial activity against Pseudomonas aeruginosa, Escherichia coli, Enterobacter aerogenes, and Streptococcus pyogenes. The main compounds responsible for this activity were flavonoids, tannins, terpenoids, saponins, and alkaloids. Both aqueous and methanol extracts showed inhibition zones against various bacteria, with different magnitudes of inhibition. The extracts from the flowers and leaves were effective against Bacillus subtilis, E. coli, and S. aureus. The antibacterial activity increased with higher extract concentrations. The identified compounds in the extracts included cardiac glycosides, anthraquinones, phlobatanins, carbohydrates, phytosterols, proteins, glycosides, tannins, phenols, and flavonoids. Alkaloids and saponins were found in both extracts.[11-15]

2. Antifungal Activity

Previous studies have confirmed the noteworthy antifungal activity of Hibiscus rosa sinensis. Its leaves, roots, and flowers yield extracts that exhibit inhibitory effects against several fungi, including Candida albicans, Aspergillus niger, and Candida parapsilosis. The antifungal properties are attributed to the presence of bioactive compounds like flavonoids, tannins, terpenoids, saponins, and alkaloids.[16]

Methanol leaf extracts of H. rosa sinensis exhibited notable inhibition against Aspergillus niger and Candida albicans. Ethyl alcohol extracts from the plant's flowers also displayed high zones of inhibition against Candida parapsilosis and Aspergillus niger.[17] Additionally, the plant's methanol extracts inhibited the growth of other fungi, including C. glabreta, A. flavus, and C. albicans, while the flower extract showed activity against various fungi except Aspergillus flavus.[18,19]

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3. Reproductive Effects

(a) Male Reproductive Effects

Benzene chloroform and alcoholic extracts decreased spermatogenic elements and sperm count in mice.[20] Ethanol, chloroform, and ethyl acetate extracts caused a decrease in testis weight and sperm parameters in mice.[21] Oral administration of aqueous and alcoholic extracts did not significantly affect the reproductive organs of male rats.[22]

(b). Female Reproductive Effects

Benzene extract disrupted the estrous cycle and reduced the weight of ovaries, uterus, and pituitary gl in rats.[23] Benzene extract caused an irregular estrous cycle, atretic follicles, and absence of corpora lutea in mice.[24] Ethanolic and benzene extracts reduced glycogen content in the uterus of rats.[25]

(c) Anti-Fertility Effects

Benzene hot extracts from flowers, leaves, and stem barks of Hibiscus rosa-sinensis prevented pregnancy in female rats.[26] Benzene extract administered during early pregnancy caused anti-implantation effects and inhibited endometrial capillaries in mice.[27] Ethanolic extract of the roots showed anti-implantation and uterotropic activity in rats.[28] Benzene extract administered during early pregnancy led to a failure of implantation and affected uterine composition in mice.[29]

(d) Effects on Diabetic Pregnancy

Aqueous extract of Hibiscus rosa-sinensis flowers had both deleterious and beneficial effects on maternal-fetal outcomes in diabetic rats.[30] Treatment with the extract improved maternal and fetal weights and reduced certain biochemical parameters in diabetic pregnant rats.[30]

4. Antioxidant Effects

Hibiscus rosa-sinensis flower extracts exhibit strong antioxidant activity. The ethanolic extract scavenged hydrogen peroxide effectively and contained alkaloids, tannins, steroids, glycosides, and flavonoids. Methanolic and ethanolic extracts showed DPPH radical scavenging activity, with the methanolic extract having higher phenolic and flavonoid contents. Aqueous stem extracts had higher DPPH scavenging activity. Water extracts of petals demonstrated significant DPPH scavenging activity due to their iron content.[31,32,33] Aqueous leaf extracts had slightly higher antioxidant activity than ethanol extracts. Different flower colors and leaf types had varying antioxidant activities. Hibiscus rosa-sinensis extract

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exhibited higher antioxidant capacity than the standard BHT. Its antioxidant activity is attributed to flavonoids and natural phenolics, making it a potential natural antioxidant source.[34,35,36,37]

5. Antidiabetic Activity

Hibiscus rosa sinensis extracts have been found to possess significant anti-diabetic properties in various animal models and clinical trials. The alcoholic leaf extracts exhibited oral hypoglycemic effects in non-obese type I diabetic mice, leading to a significant reduction in blood glucose levels.[38] Additionally, root extracts demonstrated anti-diabetic effects in alloxan-induced type II diabetic rats.[39] Furthermore, ethanolic leaf extracts showed promising results in reducing blood glucose levels and providing renal and hepatic protection in streptozotocin-induced diabetic rats.[40,41] Even the ethanolic flower extract displayed anti-hyperglycemic activity in alloxan-induced hyperlipidemic rats.[42,43,44]

In clinical trials involving type II diabetic patients, H. rosa sinensis flower powder proved beneficial by reducing fasting and postprandial blood glucose levels, as well as lowering total cholesterol and triglyceride levels.[45] These combined findings highlight the potential of Hibiscus rosa sinensis extracts as effective and multi-faceted anti-diabetic agents.

6. Antihyperlipidemic Activity

The oral administration of Hibiscus rosa-sinensis Linn flower ethanolic extract at a dosage of 500 mg/kg b.w was evaluated in hyperlipidemic rats induced by triton and atherogenic diet. In the triton model, the extract significantly lowered serum lipid levels, including total cholesterol, triglycerides, phospholipids, LDL, and VLDL, while increasing HDL levels. Similarly, in the atherogenic diet model, the extract significantly reduced serum lipid levels and increased HDL levels compared to control groups. These findings suggest the potential of Hibiscus rosa-sinensis extract in managing hyperlipidemia.[46]

7. Anticancer Activity

Hibiscus rosa-sinensis extract has been found to exhibit protective effects against tumor promotion in cancer development. In a study using mouse skin, the extract mitigated hyperproliferation and oxidative damage induced by benzoyl peroxide and ultraviolet radiation. Pretreatment with the extract restored levels of cellular protective enzymes, reduced malondialdehyde formation and hydrogen peroxide content, and decreased ornithine decarboxylase activity and thymidine incorporation in DNA.[47]

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Furthermore, the chemo-preventive activity of Hibiscus rosa-sinensis extract was investigated in mouse skin carcinogenesis induced by 7,12-dimethyl benz(a)anthracene (DMBA)/croton oil and 12-O-tetradecanoyl phorbol-13-acetate (TPA). Application of the extractbeforeo Crotonil resulted in a significant reduction in the number of tumors and the percentage of tumor-bearing mice. The extract also delayed the appearance of the first tumor. Additionally, pretreatment with the extract and the compound gentisic acid restored the levels of glutathione (GSH), antioxidant enzymes, and metabolizing enzymes, while reducing malondialdehyde formation and hydrogen peroxide content.[48]

The cytotoxic activity of different extracts of Hibiscus rosa-sinensis was evaluated against cancer cell lines. Methanol extracts of the leaf showed the highest cytotoxic activity against leukaemic cancer cells, followed by petroleum ether and ethyl acetate extracts. Similarly, chloroform and aqueous extracts of the flower exhibited inhibition of cell viability against MCF-7 cell lines in a dose-dependent manner.[49] The aqueous extracts specifically inhibited melanoma cell growth without affecting non-transformed cells and contained low molecular weight growth inhibitory compounds.[50]

8. Hair Growth Promoting Activity

Hibiscus rosa sinensis extracts have shown hair growth-promoting activity in animal studies. The petroleum ether leaf extract increased hair length and density, while the hydrolcholic leaf extract improved hair length.[51,52] The extracts stimulated the anagen phase of hair follicles and showed potential comparable to minoxidil.[53,54] Additionally, the aqueous flower extract increased hair length in rats. Further research is needed to explore their mechanisms and potential applications for hair growth.[55]

9. Antioxidant Effects

Hibiscus rosa-sinensis extracts were evaluated for antioxidant potential. Methanol and ethanol extracts showed high phenolic and flavonoid contents, as well as significant DPPH scavenging and linoleic acid inhibition.[56] Flower extracts effectively scavenged hydrogen peroxide and superoxide radicals.[57] Leaves exhibited excellent antioxidant activity surpassing synthetic antioxidants.[58] Hibiscus rosa-sinensis displayed strong antioxidant and ferric-reducing abilities.[59] Petal extracts demonstrated effective antioxidant activity in various assays.[60]

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10. Neuroprotective Activity

The methanolic extract of H. rosa-sinensis roots exhibited analgesic, sedative, and anxiolytic effects in mice and rats. It contained phytochemicals like tannins, flavonoids, saponins, and glycosides.[61] Another study showed that the methanol root extract improved learning, memory deficits, and anxiety induced by cerebral ischemia in rats.[62] The petroleum ether flower extract enhanced spatial memories in rats with cognitive decline.[63] Additionally, the ethanolic extract of H. rosa-sinensis roots displayed significant antidepressant activity in mice and rats.[64]

11. Immunomodulatory Activity

Hydro-alcoholic extract of dried Hibiscus rosa-sinensis flowers showed immunological effects, while the extract of Euphorbia neriifolia had a stronger impact on immune components.[65] The aqueous extract of Hibiscus rosa-sinensis increased plaque-forming cells, antibody titer, and DTH response in mice while altering IL-1alpha and IL-2 levels.[66]

12. Wound Healing Activity

Hibiscus rosa-sinensis flower extract demonstrated wound healing activity, reducing epithelialization time and wound area.[67] The leaf extracts also showed significant tensile strength and improved histological features in wound healing.[68] In mice, the extract enhanced wound closure and strength. It was effective in promoting healing in both thermal and chemical burn wounds.[69]

13. Gastro-protective Activity

The aqueous-ethanolic extract of Hibiscus rosa sinensis aerial parts provided mucosal protection in the gastric tissues of rats. It reduced gastric secretion and acidity while increasing pH.[70] The extract showed a significant decrease in ulcer index and exhibited gastro-protective activity in various ulcer models. The presence of carbohydrates, sterols, flavonoids, glycosides, and tannins contributed to its effects.[71]

14. Cardio-protective Activity

The administration of H. rosa-sinensis leaf extracts lowered blood pressure in both hypertensive and non-hypertensive rats. The extract showed potential kidney function interference.[72] In vitro studies demonstrated cardio-protective effects, improving post-ischemic recovery and reducing ventricular premature beats and tachycardia.[73]

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15. Anticonvulsant Activity:

The ethanolic extracts of Hibiscus rosa-sinensis flowers showed anticonvulsant activity. Further fractionation indicated that the acetone-soluble part of the ethanolic extract possessed this activity. The fraction protected animals from various types of convulsions induced by different substances, such as maximum electroshock, electrical kindling, pentylenetetrazole, lithium-pilocarpine, and electrical kindling. It also counteracted the effects of D-amphetamine and enhanced pentobarbitone-induced sleep. Additionally, it increased the levels of gamma-aminobutyric acid (GABA) and serotonin in the brain.[77]

16. Antidepressant Effects:

The crude ethanolic extract of Hibiscus rosa-sinensis flowers exhibited an antidepressant effect. The study used forced-induced swimming test (FST), tail suspension test (TST), and open field test (OFT) as parameters. The extract showed a significant dose-dependent decrease in immobility time in FST and TST. However, it did not produce significant effects on OFT. The extract also affected the levels of monoamine oxidase A (MAOA) and monoamine oxidase B (MAOB), with the 250mg/kg dose significantly influencing MAOB.[78]

17. Anti-inflammatory Activity

Hibiscus rosa-sinensis extracts have demonstrated anti-inflammatory activity in different models.

Rectal administration of H. rosa-sinensis hydroalcoholic leaf extracts reduced colon ulcer areas in rats with colitis.[79] Ethanol extracts of H. rosa-sinensis flowers and leaves reduced paw edema in rats induced by carrageenan.[80] Phytochemicals present in the extracts, such as steroids, polyphenols, alkaloids, and flavonoids, likely contribute to their anti-inflammatory effects.[79]

18. Antipyretic Activity

Hibiscus rosa-sinensis extracts showed antipyretic activity: Aqueous root extract reduced rectal temperature in rats with yeast-induced pyrexia. The extract exhibited analgesic effects in the tail-flicking test.[81] Aqueous leaf extract lowered rectal temperature in mice.[82] These findings indicate the potential of Hibiscus rosa-sinensis as an antipyretic agent.

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19. Antiparasitic Activity

Hibiscus rosa-sinensis leaf methanol extract exhibited significant antiparasitic effects against Hymenolepis diminuta.

In vitro study: The extract paralyzed the worms within 3 hours and caused mortality within 4 hours at a concentration of 40 mg/ml.

In vivo study: The highest dose (800 mg/kg) of the extract reduced eggs per gram (EPG) count by 66.55% and worm count by 75.00% in infected rats.[83]

20. Fibrinolytic Effects

The aqueous extract of Hibiscus rosa-sinensis was investigated for its fibrinolytic effect, and the specific fraction responsible for this activity was identified. The results demonstrated that the extract exhibited potent fibrinolytic activity without causing haemolysis. Among the fractions tested, fraction 5 showed the highest and most significant fibrinolytic activity.[84]

21. Anti-haemolytic Effects

The anti-haemolytic activity of Hibiscus rosa-sinensis flowers was assessed in vitro. The flower extract demonstrated a significant reduction in hydrogen peroxide-induced haemolysis and lipid peroxidation, indicating its ability to protect erythrocytes from damage.[57]

22. Urinary Effects

The extract inhibits crystal aggregation and interferes with the early stages of stone formation, suggesting potential for treating urolithiasis.[85] Urinary volume and electrolyte extraction: In rats, the extract increases urinary volume and the excretion of sodium and chloride ions.[86]

23. Hepato-protective Activity

The ethanolic extract of Hibiscus rosa-sinensis leaves alleviated the liver from piroxicam-induced toxicity in mice. The extract was administered orally at 30 mg/kg for 15 days. The liver enzymes (alanine transaminase, aspartate transaminase, and alkaline phosphatase) in the blood were significantly lower in the extract-treated group than in the piroxicam-treated group. The extract also enhanced the antioxidant status of the liver by reducing lipid peroxidation and increasing the activities of superoxide dismutase, catalase, and glutathione peroxidase, which showed its potential to prevent oxidative stress-mediated hepatic damage.[87]

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24. Effects On Colitis

Hibiscus rosa-sinensis hydroalcoholic leaf extract showed ameliorative effects in acetic acid-induced colitis in rats. It reduced oxidative stress, colonic inflammation markers (NO and TNF-α), and increased antioxidant levels (SOD and GSH) at doses of 100 and 200 mg/kg.[88]

25. Antitussive Effects

The methanolic extract of Hibiscus rosa-sinensis demonstrated antitussive activity in a cough model induced by citric acid. It significantly reduced the number of coughing episodes, similar to the effect of codeine.[89]

26. Antiviral Activity

The ethanol (80%) extract of freeze-dried plant showed equivocal antiviral activity against coxsackie B2, measles, and poliovirus I in cell culture at varying concentrations. However, it was inactive against adenovirus, herpes virus type I, and semlickiforest virus in terms of plaque inhibition.[90]

DISCUSSION

Hibiscus rosa-sinensis, also known as Japa, is a plant with ornamental value that has been grown in India since ancient times. It is widely distributed across the country and has various pharmacological activities. The plant has been described in many classical Ayurvedic texts, except the Samhitas. It can act as an antibacterial and antifungal agent, thanks to the flavonoids, tannins, terpenoids, saponins, and alkaloids in its extracts. These compounds make it a potential natural remedy for infections. Moreover, the plant can affect reproductive health, lower blood sugar and cholesterol levels, and inhibit cancer cells, suggesting a broad spectrum of therapeutic uses.

The antioxidant and neuroprotective activities of Hibiscus rosa-sinensis suggest its potential in managing oxidative stress and supporting brain health. The plant's ability to promote wound healing, gastro-protection, and cardio-protection further adds to its medicinal value. However, caution is necessary, as high doses may lead to hepatotoxicity and nephrotoxicity, emphasizing the need for proper dosage regulation and safety considerations.

CONCLUSION

In conclusion, Hibiscus rosa-sinensis exhibits diverse pharmacological activities that hold promise for natural therapeutic applications. Its extracts demonstrate potent antibacterial and

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antifungal effects, making it a potential candidate for developing antimicrobial agents. The plant's impact on reproductive health, along with its antidiabetic and antihyperlipidemic properties, suggests possible applications in managing diabetes and related cardiovascular conditions.

Moreover, Hibiscus rosa-sinensis shows promise as a potential anticancer agent, displaying chemo-preventive and cytotoxic effects against cancer cells. Its antioxidant and neuroprotective properties offer potential benefits in combating oxidative stress and supporting brain health. Furthermore, the plant's wound healing, gastro-protective, and cardio-protective activities indicate its potential in promoting tissue repair and maintaining gastrointestinal and cardiovascular health.

However, the potential for hepatotoxicity and nephrotoxicity at high doses necessitates caution in its use and highlights the importance of further research to determine safe and effective therapeutic doses.

In conclusion, Hibiscus rosa-sinensis holds great potential for natural therapeutic applications, but comprehensive research, including pre-clinical and clinical studies, is required to fully explore its medicinal benefits, identify active compounds, and ensure its safety for human use. With continued investigation, this medicinal plant may contribute significantly to the development of natural medicines for various health conditions.

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